Lecture XII - Metro Passenger Flow Control

Applied Optimization with Julia

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Introduction

FIFA World Cup 2022 in Qatar

Learning Objectives

After this lecture, you will be able to:

- **Model** time-expanded passenger flows with precomputed delays
- **Translate** a safety goal into inflow control constraints
- **Explain** how the set $\mathcal{R}_{e,t}$ connects periods and minutes
- **Embed** an optimization model into a rolling-horizon heuristic

Public transport

- FIFA World Cup 2022 took place in a small region
- The capacity of the metro system was **finite**
- More than 1 million tourists were expected
- Metro usage was **free for all ticket holders**
- Transport methods were expected to be **overloaded**

. . .

Question: **What could become an issue?**

Crowd Disasters

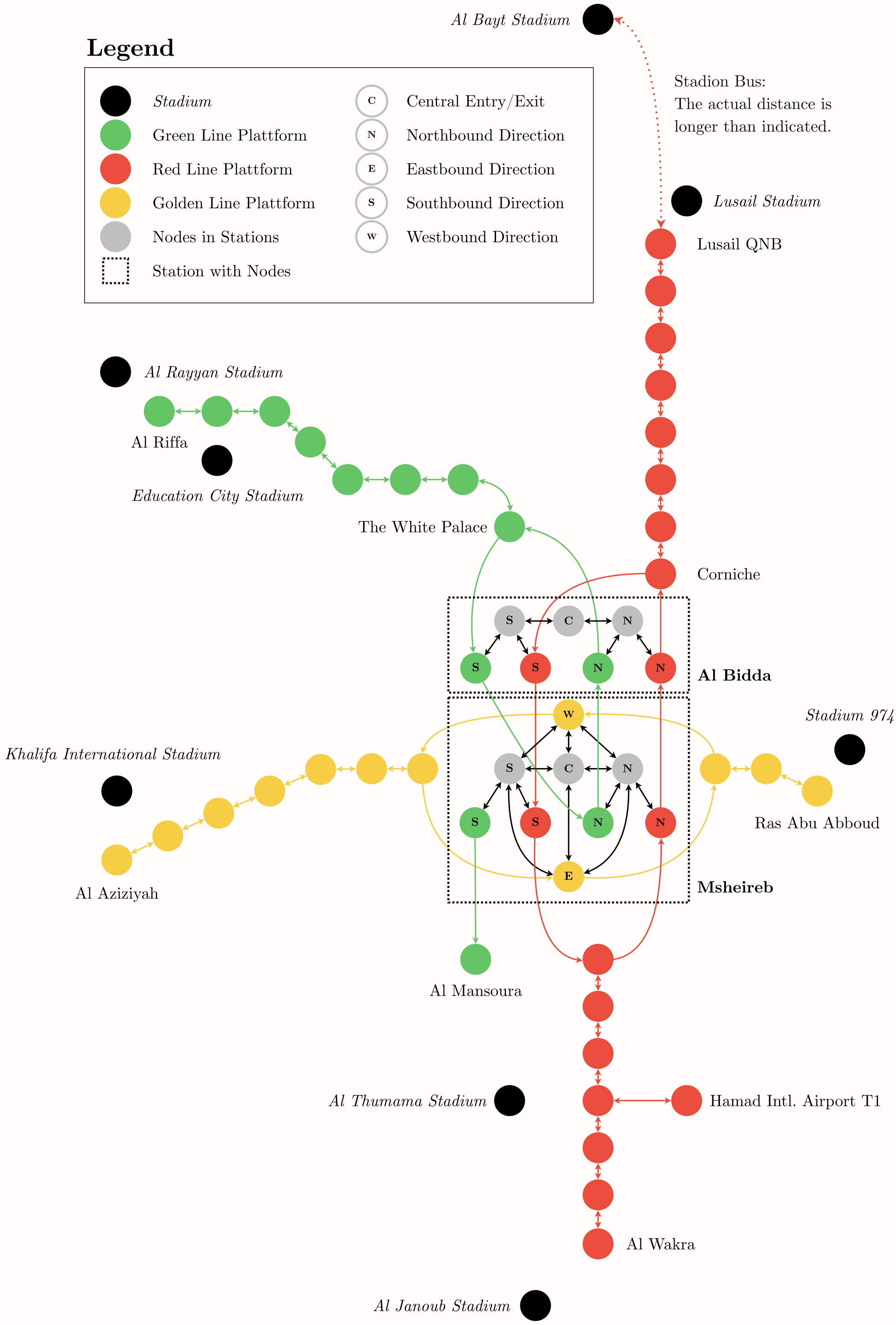
Potential locations

Question: **Where could crowd disasters happen?**

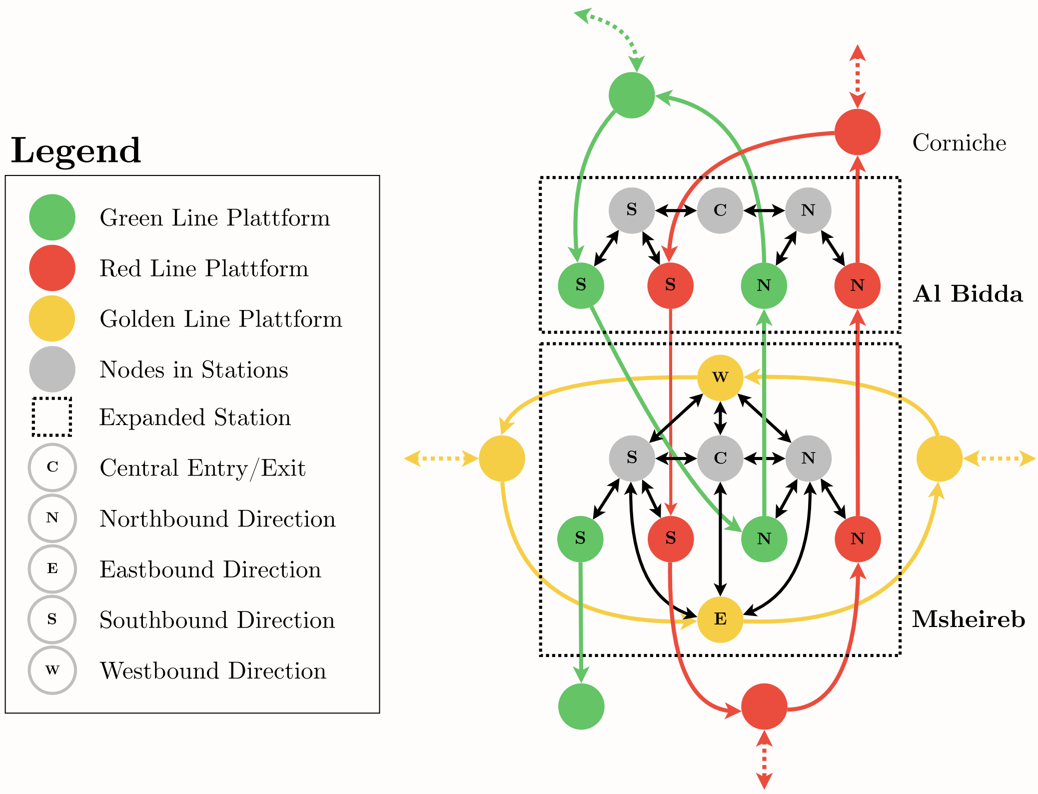
. . .

- At event venues and in waiting areas
- At intersections of multiple pedestrian flows
- In narrow passages and entrances
- On crowded metro platforms and transfer stations
- At ticket/turnstile bottlenecks and emergency exits

Metrosystem of Doha



Closer look at the Metro System



Main Bottleneck

Question: **What could be a potential bottleneck?**

. . .

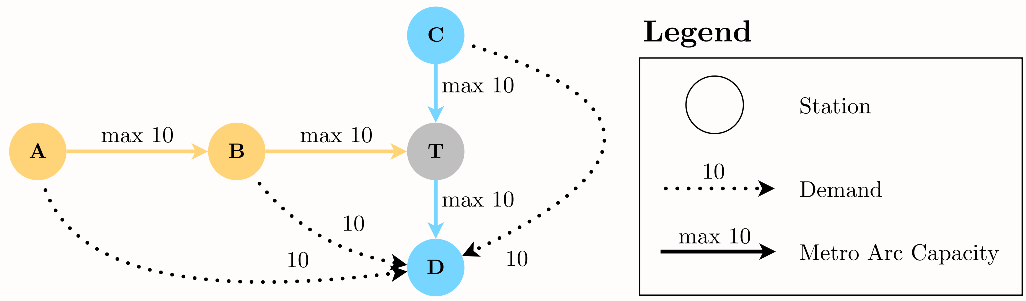
- Red Line has twice the capacity of Green and Gold Line
- People use the metro to get to the event venues

. . .

Question: **What could be the issue?**

- Imagine a match at a stadium along the Gold line
- People from Red and Green line will try to get there
- This will cause a massive crowd at transfer stations

Visualization



- 10 people want to start per minute at **each station**
- Everybody wants to get to **station D**

Capacities inside the stations

- Capacities **inside the stations** are limited as well
- These include the escalators, stairs, and elevators
- But also the platforms and the ticket gates

. . .

> ⚠ Warning
>
> This can lead to overcrowding and potential crowd disasters!

General Issues

Question: **What could also be an issue?**

. . .

- Often unknown **how many people** exactly will participate
- Gathering data is extremely important and difficult
- Crowd behavior can be unpredictable and dynamic
- Weather conditions may affect **transportation preferences**
- Cultural factors influence **crowd movement patterns**
- Emergency situations require **flexible contingency plans**

Let's take a

closer look at the

problem structure!

Problem Structure

Objective

Question: **What could be the objectives of the authorities?**

. . .

- A safe and successful event as host country
- Good publicity and a positive recognition worldwide
- Satisfied visitors that enjoyed their time

. . .

> Note: We can model none of the above directly!
>
> But we can assist in the estimation of transport demands and the design of operational plans to ensure public **safety** and a **smooth transportation** through the city.

Underlying Problem

Question: **Can we just start modeling?**

. . .

- First, we need to understand the **movement patterns**
- Event is **unprecedented**, movement patterns are unknown
- Multiple concurrent events affect flow patterns

. . .

> Tip: We need to estimate the data first!
>
> This is a **huge challenge**, but we can use **simulation** to estimate the data.

Simulation

- Based on publicly available data from the area
- Simulates **all individuals** participating at the event
- Includes **all transportation infrastructure**
- Individual mode choice based on a choice model

. . .

> Tip: Julia is a great tool for this!
>
> The simulation was built in Julia. With 1,000,000 individuals walking, using cars, buses, and the metro, simulating a day took **less than 5 minutes**.

Results of the Simulation

- Detailed movement patterns throughout the region
- Potential sections at risk in the transport infrastructure
- Potential capacity overloads at event locations

But it is still

based on a lot

of assumptions!

Main risk: Metro

Idea

Question: **What can we do to prevent crowd disasters?**

. . .

- Regulate the **inflow** at each individual station
- Ensure utilized capacity is always within bounds

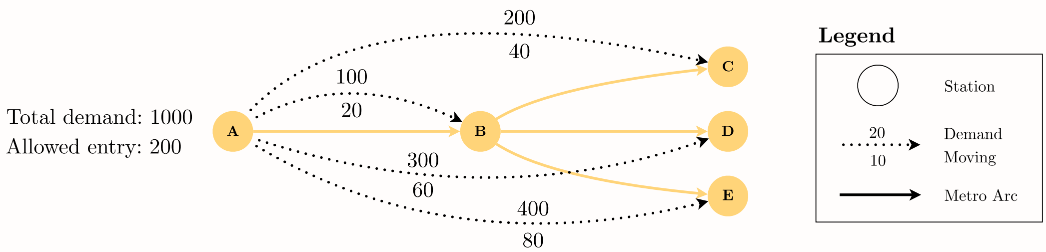
...

Question: **What could we try to model?**

...

- Minimize the queues outside of the metro stations
- Based on the **allowed inflow** and origin-destination data
- Adhering to the **capacity constraints**

Difficulty



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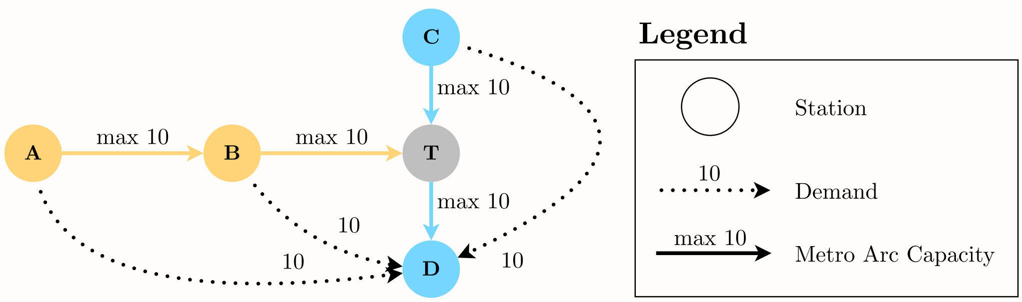
- Patterns have different origin-destination pairs
- Regulating the inflow does not affect the destination

Model Formulation

Graph Sets?

Question: **What could be the sets for the graph?**



Graph Sets?

- $\mathcal{G}$ - Connected digraph of the metro network $\mathcal{G} = (\mathcal{O}, \mathcal{E})$
- $\mathcal{O}$ - Set of metro stations, indexed by o
- $\mathcal{E}$ - Set of directed arcs between connected stations

...

ℹ Note: These are the easier sets.

These simply help us to represent the entire metro system as a graph with different nodes $\mathcal{O}$ and arcs $\mathcal{E}$ between connected stations.

Time Sets?

Question: **What could be the time sets?**

- $\mathcal{T}$ - Set of minutes in the time horizon, indexed by t
- $\mathcal{P}$ - Set of periods in the observed time horizon, where $p \in \{1, 2, ..., n\}$

...

ℹ Note: Number of periods

We further define n as the **number of periods in observed time horizon.**

Periods

Question: **Why do we add periods here?**

...

- Staff needs **clear**, consistent instructions
- Frequent changes in flow increase risk of errors
- **Easier** to manage and communicate for planners

...

ℹ Note: Period Length

We define the period length as the length of the period in minutes measured as m minutes, which is the **same for all periods**.

Mapping Minutes to Periods

- We can define an **additional set** with their relation
- It specifies the relation of periods p to minutes t.

$$I_p = \{t \in \mathcal{T} \mid (p-1) \times m + 1 \leq t \leq p \times m\} \quad \forall p \in \mathcal{P}$$

...

Question: **Can anybody explain it?**

...

- I_p is the set of minutes t that belong to period p
- Minutes are **not overlapping**, but they are **continuous**

Parameters?

Question: **What could be possible parameters?**

. . .

- $q_{o,d,p}$ - Queued passengers from station o to d with $o, d \in \mathcal{O}$ in p
- τ_e - Travel time (min) of the arcs $e \in \mathcal{E}$
- c_e - Max. allowed arc entry rate e per minute with $e \in \mathcal{E}$
- c_o^{min} - Min. station entry rate o per minute with $o \in \mathcal{O}$
- c_o^{max} - Max. station entry rate o per minute with $o \in \mathcal{O}$
- α - Safety factor applied to all capacities ($0 < \alpha < 1$)

Metro Movement

Question: **How do people move inside the metro?**

. . .

- Simple network: assume people use the **shortest path**
- Use Dijkstra's algorithm to compute it
- From each station to all other stations

. . .

> Tip: We can save the results in a new set $\mathcal{C}_{o,d}$
>
> This will help us to model the movement inside the metro.

Shortest Paths (SP)

- $\mathcal{C}_{o,d}$ - Set of arcs $e \in \mathcal{E}$ on the SP from o to d with $o, d \in \mathcal{O}$

. . .

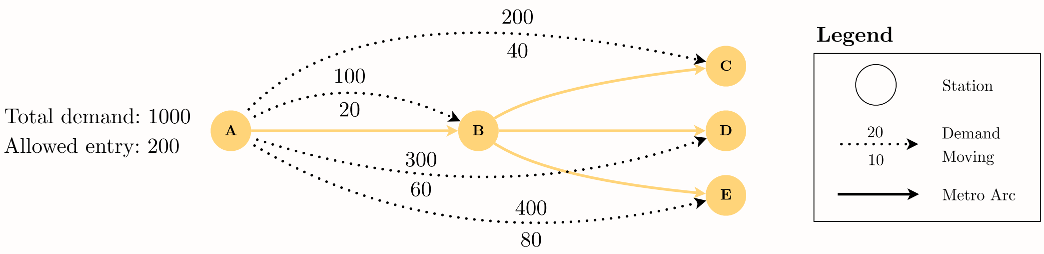
> Note
>
> The arc travel times τ_e are only the **input** to Dijkstra's algorithm. From the resulting paths we compute the travel time parameter the model actually uses:

. . .

- $\tau_{o,e}$ - Travel time (min) on the SP from station $o \in \mathcal{O}$ to arc $e \in \mathcal{E}$

People Spreading

Question: **How do people spread?**



Ratio of Origin-Destination

- We **cannot control** the destination of the passengers
- Thus we assume that people spread based on destination

...

$$\frac{q_{o,d,p}}{\sum_{f \in \mathcal{O}} q_{o,f,p}} \quad \forall o, d \in \mathcal{O}, p \in \mathcal{P}$$

...

Note

Based on the ratio of the different destinations d to the total queue for each station $o \in \mathcal{O}$ in each period $p \in \mathcal{P}$. Note that the summation needs its **own index** f over all destinations, as d is already taken by the ratio's destination.

Variables and Objective

Decision Variable?

! Our goal is to:

Regulate the **inflow** at **each individual station** to **minimize the queues** outside the metro stations based on the allowed inflow while **adhering to the capacity constraints.**

Tip

There is **one queue for all passenger flow directions**, as managing multiple queues would be too complex for the planners!

Decision Variable

i We need the following sets:

- All the metro stations, $o \in \mathcal{O}$
- All periods under observation, $p \in \mathcal{P}$

Question: **What could be our decision variable?**

...

- $X_{o,p}$ - Allowed inflow (per minute) at station o in period p

Objective Function?

! Our main objective is to:

Regulate the **inflow** at **each individual station** to **minimize the queues** outside the metro stations based on the allowed inflow while **adhering to the capacity constraints.**

...

Question: **How again are queues minimized?**

...

- By the allowed inflow $X_{o,p}$ subtracted from the queue

Objective Function

i We need the following parameters and variables:

- $q_{o,d,p}$ - People queued to travel from station o to d with $o, d \in \mathcal{O}$ in period p
- $X_{o,p}$ - Allowed inflow (per minute) at station o in period p

...

Question: **What could be our objective function?**

...

$$\text{Minimize} \quad \sum_{o \in \mathcal{O}} \sum_{p \in \mathcal{P}} (\sum_{d \in \mathcal{O}} q_{o,d,p} - m \times X_{o,p})$$

Objective Function Revisited

Question: **What does the solver really maximize here?**

...

- The queued passengers $q_{o,d,p}$ are parameters, not variables
- Their sum is a **constant** — the solver cannot change it

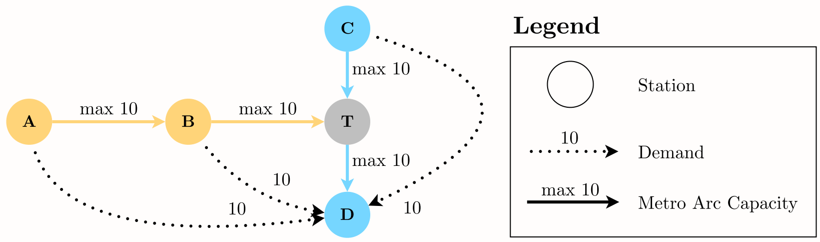
. . .

> **Note: Minimizing queues = maximizing admissions**
>
> Minimizing the objective is exactly the same as **maximizing the total admitted passengers** $\sum_{o\in\mathcal{O}}\sum_{p\in\mathcal{P}} m \times X_{o,p}$. This is also why the model gets away without explicit queue dynamics.

Constraints

Necessary Constraints

Question: **What constraints do we need?**



. . .

- The capacity of each arc is **not exceeded**
- Do not dispatch more people than are queued
- Do not dispatch less than the minimum allowed inflow

Things are going to

get a little

complicated now!

Central Question

Question: **When do people flowing into the metro system change the arcs?**

. . .

- People enter metro station at o with destination d
- They will lead to a usage of an arc e on their SP
- This usage depends on their path and the travel times

. . .

Tip: You'll likely know what we need now!

We can add a new set $\mathcal{R}_{e,t}$ to help us with this.

Set of Time-Delays

$$\mathcal{R}_{e,t} = \{(o,d,p) \mid o,d \in \mathcal{O}, p \in \mathcal{P}, q_{o,d,p} > 0, \\ e \in \mathcal{C}_{o,d}, t - \tau_{o,e} \in I_p\} \quad \forall e \in \mathcal{E}, t \in \mathcal{T}$$

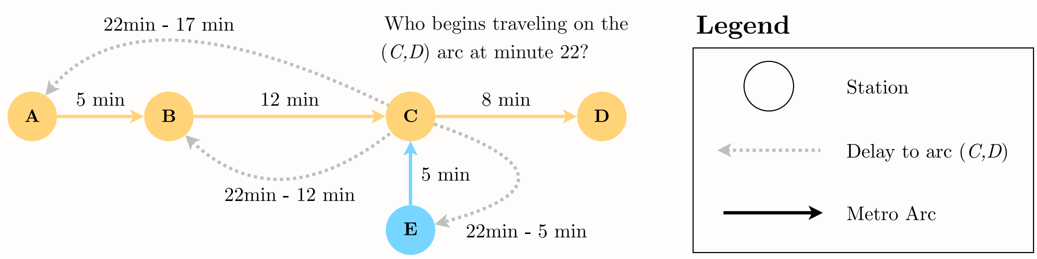
. . .

Question: **Who can explain this set?**

. . .

The set $\mathcal{R}_{e,t}$ contains **all combinations** (o,d,p) which trigger a **capacity utilization** of arc e in minute t.

Small Example



. . .

Tip

The set contains all o-d pairs and periods that result in passengers entering arc (C,D) at minute 22. With $m=2$, it is $\{(A,D,3),(B,D,5),(C,D,11),(E,D,9)\}$.

Check it yourself, e.g. from A: $\tau_{A,(C,D)} = 5+12 = 17$ and $22-17 = 5 \in I_3 = \{5,6\}$.

Essentially, we can use this set to compute the utilization of an arc at each minute based on the inflow level at the different stations and periods.

Ensure Capacity Utilization?

! The goal of this constraint is to:

Ensure that the capacity of each arc is not exceeded at any minute.

...

i We need the following variable, parameter and set:

- $q_{o,d,p}$ - people queued to travel from station $o \in \mathcal{O}$ to station $d \in \mathcal{O}$ in p
- c_e - people max. allowed to enter arc e per minute with $e \in \mathcal{E}$
- $\mathcal{R}_{e,t}$ - mapping of station entries to arc e in minute t with $e \in \mathcal{E}$ and $t \in \mathcal{T}$
- α - safety factor applied to all capacities $(0 < \alpha < 1)$
- $X_{o,p}$ - the allowed inflow per minute at metro station o in the period p

Ensure Capacity Utilization

Question: **What could be the constraint?**

...

$$\sum_{(o,d,p) \in \mathcal{R}_{e,t}} X_{o,p} \times \frac{q_{o,d,p}}{\sum_{f \in \mathcal{O}} q_{o,f,p}} \leq \alpha \times c_e \quad \forall e \in \mathcal{E}, t \in \mathcal{T}$$

...

i Note

Here, we combine the inflow at each station based on the o-d proportion and let the people spread through the network, checking that no arc is over-utilized at any minute.

Bound the Inflow Rate?

! The goal of this constraint is to:

Ensure that the inflow rate at each station stays within operational bounds.

...

i We need the following variables and parameters:

- $X_{o,p}$ - Allowed inflow (per minute) at station o in period p
- c_o^{min} - Min. station entry rate o per minute
- c_o^{max} - Max. station entry rate o per minute
- α - Safety factor applied to all capacities $(0 < \alpha < 1)$

Bound the Inflow Rate

Question: **What could be the constraint?**

. . .

$$c_o^{min} \leq X_{o,p} \leq \alpha \times c_o^{max} \quad \forall o \in \mathcal{O}, p \in \mathcal{P}$$

. . .

ℹ Note

This constraint bounds the inflow rate at each station between the **minimum and maximum** allowed entry rates, ensuring compliance with **operational capacity limits**.

Dispatch Only Available People?

! The goal of this constraint is to:

Ensure that we do not dispatch more people than are queued.

. . .

ℹ We need the following variables and parameters:

- $X_{o,p}$ - Allowed inflow (per minute) at station o in period p
- $q_{o,d,p}$ - People queued to travel from station o to d with $o, d \in \mathcal{O}$ in p
- m - period length in minutes

Dispatch Only Available People

Question: **What could be the constraint?**

. . .

$$\sum_{d \in \mathcal{O}} q_{o,d,p} - m \times X_{o,p} \geq 0 \quad \forall o \in \mathcal{O}, p \in \mathcal{P}$$

. . .

ℹ Note

This constraint ensures **non-negative remaining queues**, preventing the solver from dispatching more passengers than available in the queue.

A Hidden Pitfall

Question: **What if a queue is smaller than $m \times c_o^{min}$?**

. . .

- The lower bound forces $m \times X_{o,p} \geq m \times c_o^{min}$
- But we may only dispatch up to $\sum_{d \in \mathcal{O}} q_{o,d,p}$ people
- Example: $c_o^{min} = 2$, $m = 15$, but only 20 people queued
- A quiet station late at night makes the model infeasible!

. . .

Question: **How could we fix this?**

. . .

- Set $c_o^{min} = 0$, cap the bound at the queue, or soften the constraint

Metro Inflow Model

subject to:

Model Characteristics

Characteristics

Questions: **On model characteristics**

- Is the model formulation linear/ non-linear?
- What kind of variable domains do we have?

. . .

> Note: The answers
>
> It is a linear program, as objective and constraints are linear. The variable $X_{o,p} \geq 0$ is **continuous** — fractional inflow rates per minute are perfectly acceptable here.

Model Assumptions

Questions: **On model assumptions**

- What assumptions have we made?
- What are likely issues that can arise if applied?
- Have we thought in detail about queues?
- Are shortest paths a feasible assumption?

Implementation and Impact

Metro Inflow Optimization

. . .

Question: **Can this be applied?**

Metro Inflow Problem

- Solved very fast within seconds for realistic problem sizes
- **But** we cannot plan or control the metro inflow

- Queues are **too simplified** with passengers disappearing

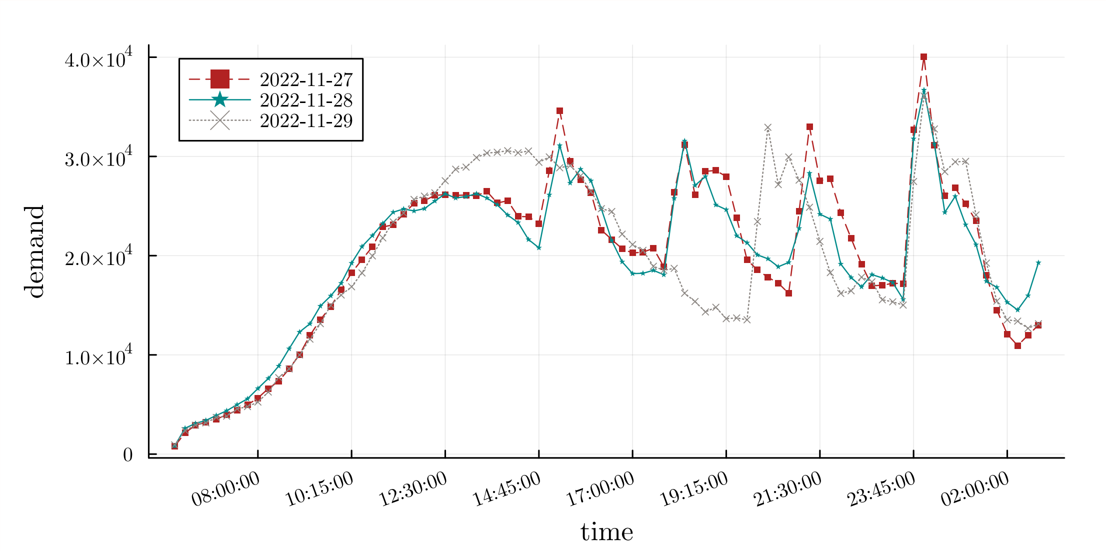
. . .

Question: **Any ideas, how the current model could be improved or how it could be embedded into a heuristic?**

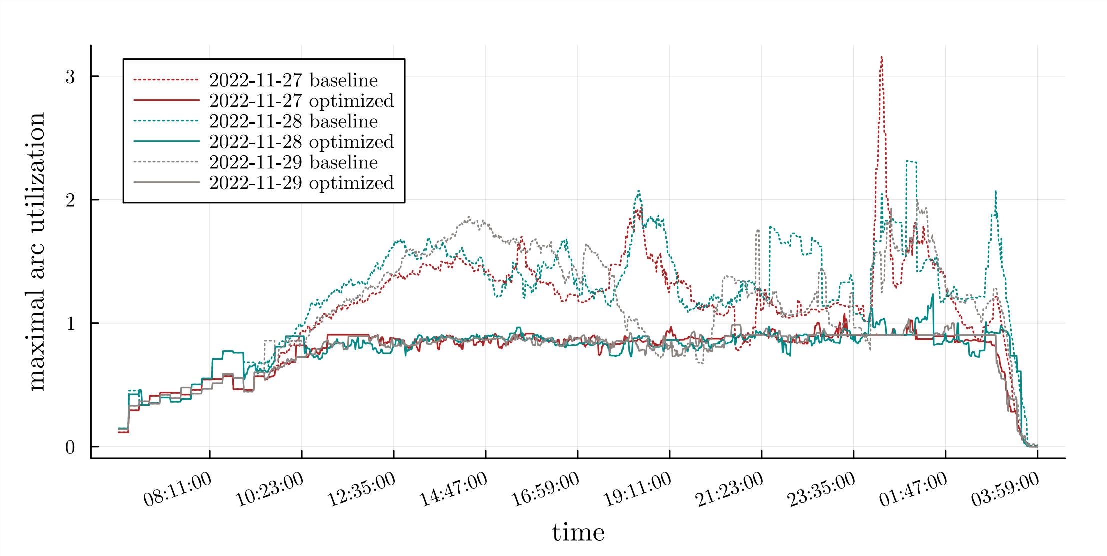
Heuristic: Step-Wise

- Solve the model for the time-horizon of a **few periods**
- Fix the inflow in the **current first period**
- Decrease capacity in the network **based on the inflow**
- Transfer **remaining queues** into the subsequent period
- Solve the model again
- **Repeat**, until the inflow is computed for all periods

Transport Demand



Utilization Analysis



Key Results

- Model reduces capacity violations from 25,941 to just 4
- Maximum arc utilization drops from **316% to 109%**
- 81% of passengers accommodated safely
- Computation time: **~1.4 seconds** per planning period

. . .

> **i** Note: Safety vs. Throughput Tradeoff
>
> The remaining 19% of passengers experience external queuing - this is the price for preventing dangerous overcrowding inside the system.

Didn't We Bound the Utilization?

Question: **Our constraint enforces $\alpha \times c_e$ with $\alpha < 1$. How can utilization still reach 109%?**

. . .

- The constraint binds the model for the forecast demand
- The 109% occurs in the **rolling-horizon simulation** with realized demand
- Realized flows can deviate from the forecast — but far less than **316%**

Scalability: Shanghai Metro

- Validated on **302 stations** (~8× more than Doha's 37)
- 99% reduction in capacity violations
- Computation: **< 2 minutes** per period
- Works for routine **peak-hour demand**

. . .

Tip: Generalizable Approach

The method scales from event-driven demand (Qatar World Cup) to routine metropolitan operations (Shanghai rush hour).

Implementation

- **Assumption of known destinations** is strong
- Movements **seemed to follow our forecasts**
- We did achieve our goal of **metro inflow control**
- Simulation was used to **estimate the inflows**

. . .

Note

Few dangerous situations, especially at the FIFA Fan Fest, were **handled well** by the authorities.

Comparison with Others

Feature	Other Methods	Our Approach
FIFO Enforcement	Rarely	Yes
Single Queue	Often separate	Yes
Real-Time Capable	Some	Yes (seconds to minutes)
Network Scale	Up to 327 stations	Comparable (302 stations)

. . .

Note: State-of-the-Art

First network-level approach combining FIFO enforcement, single-queue modeling, and real-time computational performance.

Wrap Up

- Model can help to achieve a good balance
- Can be adapted easily to **any metro system worldwide**
- Especially interesting for **larger Asian cities**

. . .

And that's it for today's lecture!

We now have covered a metro inflow control problem based on a real-world application and are ready to start solving some new tasks in the upcoming tutorial.

Questions?

Literature

Literature I

For more interesting literature to learn more about Julia, take a look at the literature list of this course.

Bibliography